

Customer Integration Guide

SIP Registration over WebSocket (WSS)

CloudConnect

1. Overview

This document describes the connection and SIP registration flow for establishing a SIP client connection with CloudConnect using WebSocket Secure (WSS). SIP signalling is transported over a secure WebSocket connection.

Connection Details

Parameter	Value
WebSocket Protocol	WSS
WebSocket URL	wss://sip2.cloud-connect.in:7443/
SIP Domain / Registrar	sip2.cloud-connect.in
SIP Port	7065
Transport	WebSocket Secure (WSS)

2. WebSocket Connection Flow

The connection follows this sequence:

- SIP Client / Web Softphone → Establish WSS connection
- WSS URL: wss://sip2.cloud-connect.in:7443/
- CloudConnect WSS Service → SIP REGISTER
- CloudConnect → 401 Unauthorized (Digest Authentication Challenge)
- Client → REGISTER with Authentication
- CloudConnect → 200 OK
- SIP Registration Successful

3. Step 1 – Establish WSS Connection

The SIP client first establishes a secure WebSocket connection with CloudConnect.

wss://sip2.cloud-connect.in:7443/

The WebSocket connection is responsible for transporting SIP signalling between the client and CloudConnect. The client must be able to reach TCP port 7443 over the network/firewall.

4. Step 2 – Send SIP REGISTER

Once the WebSocket connection is successfully established, the client sends a SIP REGISTER request. The SIP registration is associated with the CloudConnect SIP domain:

sip:sip2.cloud-connect.in:7065

5. Step 3 – Authentication Challenge

CloudConnect responds with SIP/2.0 401 Unauthorized. This can be an expected part of the SIP Digest Authentication process rather than a configuration failure.

The response contains an authentication challenge, including parameters such as:

- realm
- nonce

The SIP client uses these parameters together with the configured username and password to generate the Digest Authentication response.

6. Step 4 – Authenticated REGISTER

The client then sends another REGISTER request containing the required Digest Authentication information.

REGISTER + Authorization: Digest

- username
- realm
- nonce
- uri
- response

7. Step 5 – Successful Registration

If the credentials and SIP configuration are valid, CloudConnect responds with:

SIP/2.0 200 OK

This confirms that the SIP registration has been successfully established. The client can then proceed with SIP operations such as making and receiving calls, subject to the configured account permissions.

8. Complete Registration Sequence

- Client → WSS Connection → CloudConnect
- wss://sip2.cloud-connect.in:7443/
- Client → SIP REGISTER → CloudConnect
- CloudConnect → 401 Unauthorized → Client
- Client → REGISTER + Authorization → CloudConnect
- CloudConnect → 200 OK → Client
- Registration Successful

9. Important Port / Transport Details

WebSocket Transport

URL: wss://sip2.cloud-connect.in:7443/

Port: 7443

This is used to establish the secure WebSocket connection.

SIP Service

SIP Service: sip:sip2.cloud-connect.in:7065

Port: 7065

This represents the SIP service/domain used by the SIP signalling.

The WebSocket transport and SIP service ports should not be confused. The WebSocket connection should use wss://sip2.cloud-connect.in:7443/.

10. Troubleshooting

A. WSS Connection Not Established

- DNS resolution for sip2.cloud-connect.in
- Network connectivity
- Firewall rules
- TCP port 7443
- TLS certificate validation
- Browser/client WebSocket support

In a browser-based client, use Developer Tools → Network → WS to verify the WebSocket connection. The expected connection is:

wss://sip2.cloud-connect.in:7443/

B. 401 Unauthorized

A 401 Unauthorized response during the initial REGISTER can be expected because CloudConnect uses SIP Digest Authentication. The key check is whether the subsequent authenticated REGISTER receives 200 OK.

- Username
- Password
- Realm
- Authentication username
- SIP domain
- Digest calculation
- Client SIP stack configuration

C. 403 Forbidden

A 403 Forbidden response indicates that the request was rejected by the server.

- SIP account configuration
- Source IP/network restrictions, if applicable
- SIP credentials
- Domain/realm configuration
- User provisioning
- Server-side SIP logs

D. 405 Method Not Allowed

If the server returns 405 Method Not Allowed, verify that the SIP client is using supported SIP methods and that the request is being sent to the correct SIP service.

E. Incorrect WebSocket URL

Correct:

wss://sip2.cloud-connect.in:7443/

Examples of incorrect combinations:

- ws://sip2.cloud-connect.in:7065/
- wss://sip2.cloud-connect.in:7065/

The WebSocket transport and SIP service ports serve different purposes.

11. Validation Checklist

- DNS resolution of sip2.cloud-connect.in
- TCP connectivity to port 7443

- WSS connection successfully established
- SIP REGISTER generated by client
- Initial 401 Unauthorized received
- Authenticated REGISTER generated
- Authentication credentials verified
- Authenticated REGISTER receives 200 OK
- SIP account provisioned correctly
- No firewall/TLS issue observed
- Client is using the correct WSS URL

12. Expected Successful Flow

